

 [View source on GitHub \(https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/math_ops.py#L938-L1020\)](https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/math_ops.py#L938-L1020)

Casts a tensor to a new type.

```
tf.cast(  
    x, dtype, name=None  
)
```

Used in the notebooks

Used in the guide	Used in the tutorials
• Multilayer perceptrons for digit recognition with Core APIs (https://www.tensorflow.org/guide/core/mlp_core) • Logistic regression for binary classification with Core APIs (https://www.tensorflow.org/guide/core/logistic_regression_core) • tf.data: Build TensorFlow input pipelines (https://www.tensorflow.org/guide/data) • Distributed training with Core APIs and DTensor (https://www.tensorflow.org/guide/core/distribution) • Import a JAX model using JAX2TF (https://www.tensorflow.org/guide/jax2tf)	• Scalable model compression (https://www.tensorflow.org/tutorials/optimization/compression) • DeepDream (https://www.tensorflow.org/tutorials/generative/deepdream) • Parameter server training with ParameterServerStrategy (https://www.tensorflow.org/tutorials/distribute/parameter_server_training) • Learned data compression (https://www.tensorflow.org/tutorials/generative/data_compression) • Neural style transfer (https://www.tensorflow.org/tutorials/generative/style_transfer)

The operation casts `x` (in case of `Tensor`) or `x.values` (in case of `SparseTensor` or `IndexedSlices`) to `dtype`.

For example:

```
>>> x = tf.constant([1.8, 2.2], dtype=tf.float32)  
>>> tf.cast(x, tf.int32)  
<tf.Tensor: shape=(2,), dtype=int32, numpy=array([1, 2], dtype=int32)>
```

Notice `tf.cast` (https://www.tensorflow.org/api_docs/python/tf/cast) has an alias `tf.dtypes.cast` (https://www.tensorflow.org/api_docs/python/tf/cast):

```
>>> x = tf.constant([1.8, 2.2], dtype=tf.float32)  
>>> tf.dtypes.cast(x, tf.int32)  
<tf.Tensor: shape=(2,), dtype=int32, numpy=array([1, 2], dtype=int32)>
```

The operation supports data types (for `x` and `dtype`) of `uint8`, `uint16`, `uint32`, `uint64`, `int8`, `int16`, `int32`, `int64`, `float16`, `float32`, `float64`, `complex64`, `complex128`, `bfloat16`. In case of casting from complex types (`complex64`, `complex128`) to real types, only the real part of `x` is returned. In case of casting from real types to complex types (`complex64`, `complex128`), the imaginary part of the returned value is set to `0`. The handling of complex types here matches the behavior of `numpy`.

Note casting nan and inf values to integral types has undefined behavior.

Note this operation can lead to a loss of precision when converting native Python `float` and `complex` variables to `tf.float64` (https://www.tensorflow.org/api_docs/python/tf#float64) or `tf.complex128` (https://www.tensorflow.org/api_docs/python/tf#complex128) tensors, since the input is first converted to the `float32` data type and then widened. It is recommended to use `tf.convert_to_tensor` (https://www.tensorflow.org/api_docs/python/tf/convert_to_tensor) instead of `tf.cast` (https://www.tensorflow.org/api_docs/python/tf/cast) for any non-tensor inputs.



Args	
x	A <code>Tensor</code> or <code>SparseTensor</code> or <code>IndexedSlices</code> of numeric type. It could be <code>uint8</code> , <code>uint16</code> , <code>uint32</code> , <code>uint64</code> , <code>int8</code> , <code>int16</code> , <code>int32</code> , <code>int64</code> , <code>float16</code> , <code>float32</code> , <code>float64</code> , <code>complex64</code> , <code>complex128</code> , <code>bfloat16</code> .
dtype	The destination type. The list of supported dtypes is the same as <code>x</code> .
name	A name for the operation (optional).

Returns	
A <code>Tensor</code> or <code>SparseTensor</code> or <code>IndexedSlices</code> with same shape as <code>x</code> and same type as <code>dtype</code> .	
Raises	
<code>TypeError</code>	If <code>x</code> cannot be cast to the <code>dtype</code> .

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